Next Generation IP: IPv6 Architecture

Coverage: IPv6 128-Bit Addressing, 40-Byte Fixed Header, Extension Headers, IPv4 vs IPv6 Comparison, Transition (Dual Stack, Tunneling)

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1. IPv6 Header & Improvements

- 128-bit addresses in colon-hex notation. Fixed 40-byte base header (no checksum, no in-network fragmentation).
- Extension headers daisy-chained via Next Header field.

2. Transition Strategies

- Dual Stack, Tunneling (6to4, Teredo), NAT64/DNS64 header translation.

■ KTU University High-Yield Exam Questions

- Q: Compare IPv4 and IPv6 header structures and transition mechanisms. (8 Marks)